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Multisensory Interactions in Greenway Plazas of Differing Openness and Effects on User Behaviors

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Abstract

Spatial openness affects the subjective evaluation of soundscape, landscape, and thermal perceptions, leading to various restoration effects and recreational behaviors. However, the literature lacks studies investigating the effects of multisensory interactions under different levels of spatial openness in plazas on users' behaviors in urban greenways. Thus, this study contributes to the enhancement of recreational experiences and the environmental design of urban greenways by examining the interaction between multisensory evaluations and recreational behaviors in greenway plazas with different levels of spatial openness. Three types of plazas (enclosed, semi-enclosed, open) were selected along an urban greenway to analyze interactions through in situ measurements, questionnaires, and behavior observation. The results showed that people rated the environment as the quietest and coolest in enclosed plazas, although the sound pressure level of these plazas was the highest. Furthermore, the visual evaluation (VE) was mostly correlated with acoustic evaluation (AE) in plazas with high openness, while the correlation effect between AE and thermal evaluation (TE) was only significant in enclosed plazas. In other words, AE was the key factor targeting the improvement in comfort in greenway plazas. Secondly, improving AE was more effective for stimulating the frequency of interactive activities in enclosed plazas, compared to improving TE. However, AE had a negative effect on the time that people were willing to spend on interactive activities in semi-enclosed plazas. Finally, these findings provide corresponding strategies for creating comfortable audio, visual, and thermal environments in greenway plazas with different levels of openness, as well as strategies for enhancing the recreational experiences of visitors.

Keywords: acoustic comfort; landscape openness; outdoor activities; thermal comfort; urban greenspace

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1. Introduction

With the acceleration of urbanization and people's rising demand for natural green spaces in urban areas, greenways not only provide recreational spaces for various activities, such as exercising, sightseeing, and social interaction, but they also purify the hazardous matter in the city air, creating an eco-friendly environment. The greenway connects natural and artificial landscapes, such as forests, wetland parks, mountains, and cultural plazas; it provides a linear space for people to be with nature and satisfy different activity needs [1–3]. Connected natural areas decrease the density of particulate matter, providing natural

solutions for improving the air quality of greenways [4,5]. Plazas are important nodes along a linear greenway; they are spaces where people can rest and carry out leisure activities along the greenway. Many studies revealed the restoration effects of such plazas—those that had a high percentage of green vegetation and sky visibility, coupled with natural sounds, such as birds and wind, induced restoration in people and promoted recreational behaviors [6–8]. However, a thermal evaluation significantly influenced the above effect, especially during the hot summer climate, which led to negative environmental satisfaction and prevented people from participating in activities. In contrast, an increasing number of studies started to focus on the integrated effect of physical environments, such as the effect of spatial openness on visual, acoustic, and thermal evaluation, and users' behaviors in greenway plazas.

Previous studies found that spatial openness had a significant effect on the physical environment, subjective evaluation, and emotional response. The spatial openness of a greenway was defined as the visibility of landscape elements—such as water, sky, and vegetation—as well as whether it was affected by surrounding green or grey infrastructures—like trees and facilities. It was proven that spatial openness influenced visual, acoustic, and thermal evaluations; different perspectives of the above evaluations also influenced each other as multisensory evaluations. For visual and acoustic perceptions, the densities of trees and enclosing vegetation influenced spatial openness and sound pressure levels, which further influenced people's tranquility and overall comfort [9–11]. The users of greenways, including cyclists and walkers, preferred views with high levels of openness. Cyclists were also affected by changing patterns in the landscape when moving positions [12–16]. On the other hand, water and bird sounds, coupled with high openness and natural views, made people feel at ease more easily, compared to the effect of traffic noise [17,18]. As for acoustic and thermal perceptions, cyclists and walkers both felt annoyed by traffic noise in spaces with high openness and felt relieved by natural sounds, such as water and bird sounds, in greenways [19,20]. Traffic noise with high sound pressure levels resulted in high thermal sensitivity, while the natural soundscape reduced the impact of traffic noise and enhanced thermal comfort [21,22]. Therefore, the spatial openness of greenways influenced the percentage of natural views, sound pressure levels, and other physical indicators, leading to varying multisensory evaluations. However, fewer studies deeply analyzed the feature of multisensory evaluations at greenway plazas with different levels of spatial openness and the relationship between spatial openness and multisensory evaluations.

In the process of spatial openness affecting multisensory evaluations, the interactive effect between each evaluation also plays an important role by influencing the overall satisfaction of the environment. For visual and acoustic evaluations, previous studies indicated that high openness views of natural greenways with attractive elements could improve the landscape aesthetic, thus enhancing the soundscape pleasantness and harmonization [23–25]. Conversely, the enclosed spaces of greenways improved tranquility and relieved mental stress [26–28]. Regarding visual and thermal evaluations, high openness and a high percentage of green coverage were beneficial for visual comfort and decreasing thermal sensitivity, which further improved the overall comfort [29,30]. From the perspective of acoustic and thermal evaluations, the annoyance of traffic noise in the open space of greenways resulted in high thermal sensitivity, while the natural soundscape reduced the negative impact of traffic noise [31–33]. Overall, the open natural space of greenways assisted people in generating positive feelings; in addition, to endure the impact of high temperatures and traffic noise, the enclosed space of greenways provided a tranquil space for people with less soundscape eventfulness. However, fewer studies focused on the

overall interactive effect between visual, acoustic, and thermal evaluations as multisensory interactions at different levels of spatial openness of greenway plazas.

While the spatial openness of greenway plazas affects the interaction effect of multisensory evaluations, it further influences users' behaviors. In terms of visual and acoustic perceptions, previous studies revealed that a high openness view coupled with natural sounds, such as bird sounds, significantly improved the pleasantness and acoustic comfort of people, which further increased the proportion of people participating in social activities [34,35]. Other studies indicated that exercisers preferred natural views along greenways, while visitors to greenways preferred to stay at public facilities and enjoy the lake view [36,37]. For thermal and acoustic evaluations, an open space with a high level of traffic noise and low density of trees reduced the walking speed of visitors, while bird sounds with pleasantness attracted people to head in the direction of sound sources [38,39]. Overall, current studies mainly focused on investigating the impact of visual, acoustic, and thermal perceptions on users' behaviors separately, and fewer studies paid attention to analyzing the effect of multisensory interactions on their behaviors at different levels of spatial openness of greenway plazas.

In summary, the spatial openness of greenway plazas affected multisensory evaluations, interactions between multisensory evaluations from visual, acoustic, and thermal perceptions, and users' behaviors. Fewer studies comprehensively investigate the above process (Figure 1). In order to bridge these gaps, this study aimed to explore the interactive effect between spatial openness, multisensory interactions, and recreational behaviors in greenway plazas. To achieve this goal, this study put forward two key scientific questions:

Research Question 1 (RQ1): What are the characteristics of the physical environment, multisensory interactions of visual, acoustic, and thermal perceptions, and users' behaviors at greenway plazas with different levels of spatial openness?

Research Question 2 (RQ2): How do the spatial openness and multisensory interactions of greenway plazas influence users' behaviors?

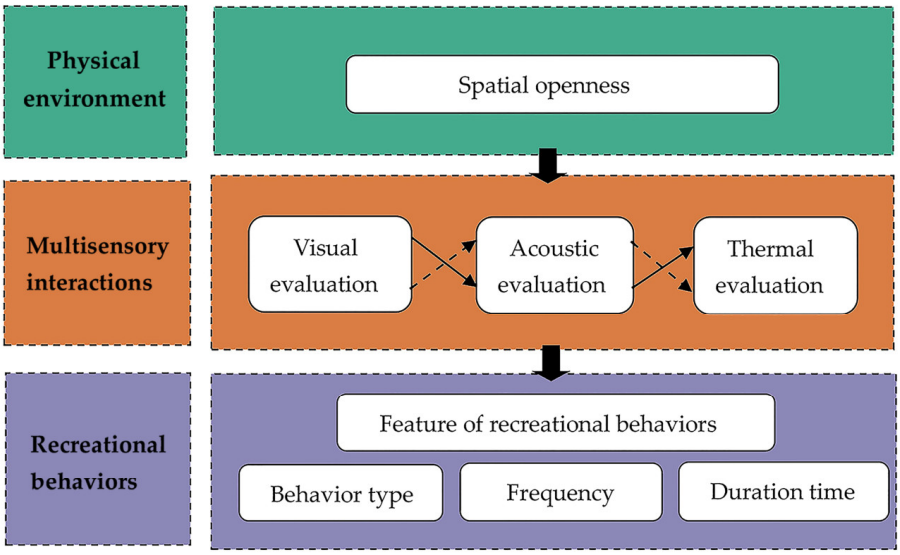


Figure 1. Conceptual framework of the relationship between spatial openness, multisensory interactions, and recreational behaviors at greenway plazas. The continued and dashed arrows in the multisensory interaction part represent the interactive impact between visual, acoustic, and thermal evaluations.

To answer these questions, this study selected the East Lake greenway, Wuhan, China, with a typical hot summer climate as the study area. Three types of spatial openness (enclosed, semi-enclosed, and open) of the greenway plazas were chosen to compare the

differences in the interaction effect of visual, acoustic, and thermal perceptions, as well as investigate the effect of multisensory interactions on users' behaviors. Overall, this study provided data support and strategies for designing the landscape, soundscape, and thermal environment of the greenway plaza while enhancing recreational experiences and positive user behaviors.

2. Materials and Methods

2.1. Studied Area

Wuhan has a typical summer-hot and winter-cold climate with extreme hot temperatures of up to 40.5 °C lasting 50 days in the central districts. With the acceleration of urbanization in Wuhan, the landscape of Wuhan is facing challenges, such as a lack of specificity and degradation of biological diversity. To address the rising demand for natural and green spaces and cope with high temperatures, governments have constructed 29 various types of greenway walking routes, connecting different types of landscape spaces, including wetland parks, forests, water bodies, and mountains. Each walking route has a unique topic with specific landscape elements and changing patterns, such as mountain forest lines and forests attached to lake lines. The natural and artificial elements, such as trees, plants, rocks, historical buildings, and pavilions, vary according to the type of land. Moreover, the natural parks, forests, and wetlands along greenways are rich in natural sound sources, while squares and historic parks are dominated by the sound of human activities. Since bikes and electronic cars are available on the main greenway, traffic noise, such as honking and braking, affects the acoustic environment of the total greenway.

As illustrated in Figure 2, this study selected the walking route from the north entrance (Liyuan Square) to the south exit (Yikeshu) of the greenway as the studied route with a high density of people, according to their walking patterns and favorable lines. According to previous studies, the spatial openness of the greenway plaza could be classified according to the percentage of the sky visibility ratio [40]. The measuring points in greenway plazas were selected based on the spatial openness and the type of enclosure structure. Six measuring points were chosen and divided into three categories (enclosed, semi-enclosed, and open). Measuring points R1 and R4 were enclosed plazas with the largest amount of green vegetation and historical buildings. People usually stayed and enjoyed the tranquil and historical atmosphere. Measuring points R2 and R3 were semi-enclosed plazas surrounded by modern structures and natural elements, such as trees, shrubs, and flowers. People gathered here to listen to concerts or catch fish. Regarding points R5 and R6, the plazas were open plazas with a broad view and less green vegetation. People might stay for a while to enjoy the lake view.

2.2. In Situ Measurement of the Feature of the Visual, Acoustic, and Thermal Environment

As the urban densification in Wuhan accelerated, the average air temperature increased by 1.3 degrees from 2008 to 2018 [41]. In order to objectively evaluate the physical environment, the percentage of visual elements, L_{Aeq_5min}, air temperature, and relative humidity were selected as the physical parameters in this study. For the visual environment, panoramic images were taken by a digital camera (Sony Alpha 7R V, Sony, Tokyo, Japan) at a height of 1.5 m above ground in the time period of 09:00–10:00 a.m. to identify the spatial openness of each measuring point. The shooting point was at the center of the sample square, where people gathered for activities. Then, the percentage of sky elements was calculated based on the panoramic images using the following equation:

$$S_i = P_s/P \times 100\% \quad (1)$$

where P_s represents the area coverage of sky elements in the panoramic image, P indicates the whole visibility area in the panoramic image, and S_i shows the percentage of sky elements in the whole visibility area of human beings as the spatial openness. After the calculation, the spatial openness was divided into three categories: enclosed (up to 20%), semi-enclosed (20–40%), and open (more than 40%) ranges (Table 1).

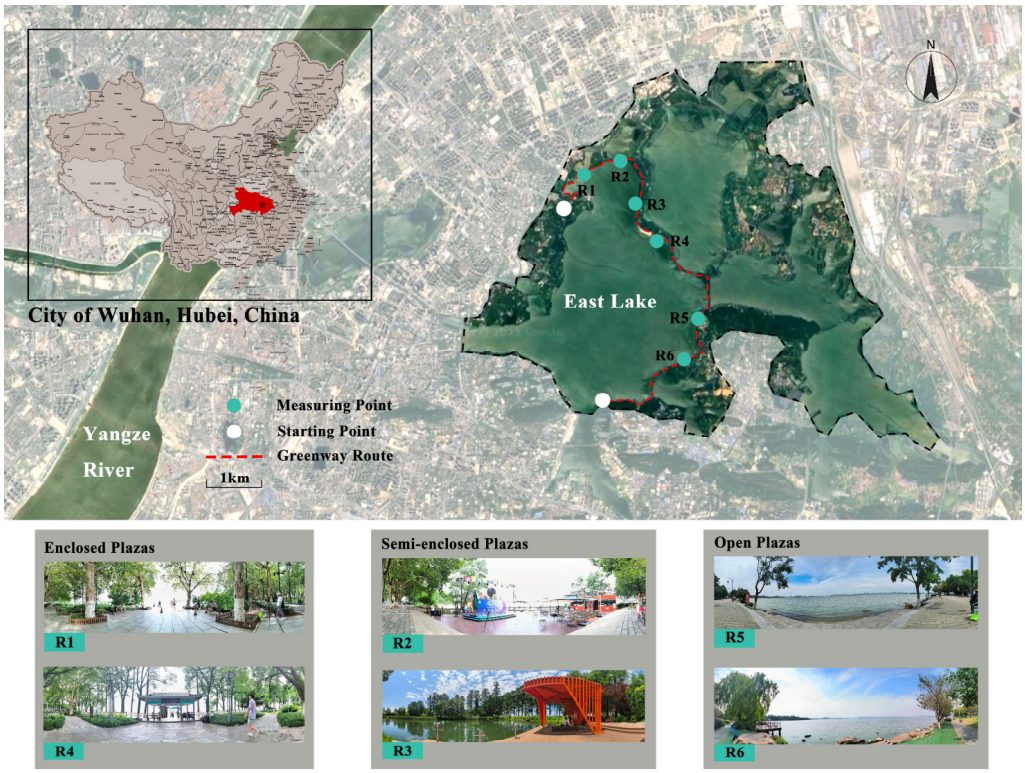


Figure 2. Layout of visitors' walking routes in the East Lake greenway in Wuhan, China, and six measuring points at the plaza with different levels of spatial openness along the greenway. The red area represents the province of the study site. The measuring points are as follows: R1, lake shore square; R2, musical square; R3, wetland park square; R4, ancient pavilion square; R5, rock island square; R6, eyes of lake square.

Table 1. Area coverage, spatial openness, and percentage of sky ratio of each greenway plaza at each measuring point.

Measuring Point	Covering Area/m^2	Spatial Openness	Percentage of Sky Visibility Ratio/%
R1	246	Enclosed	Up to 20%
R2	268	Semi-enclosed	20–40%
R3	241	Semi-enclosed	20–40%
R4	310	Enclosed	Up to 20%
R5	302	Open	More than 40%
R6	290	Open	More than 40%

For the acoustic environment, L_{Aeq_5min} was measured by an Aihua AWA 6228+ sound level meter (Hangzhou Aihua Intelligent Technology, Hangzhou, China) at a distance of 1.2 m away from the reflecting surface continuously for 5 min. The sound level meter measured the instant sound pressure level at each second, and L_{Aeq_5min} was calculated using the equation:

$$L_{Aeq,T} = 10\log_{10}((\sum 10^{(L_{A,i}/10)})/N) \quad (2)$$

where $L_{A,i}$ represents the instant sound pressure level at each second. N shows the number of times that $L_{A,i}$ was measured. $L_{Aeq,T}$ indicates the equivalent sound pressure level when T is 5 min. During the measurement, the ambient sound of each measuring point was also recorded by a Sony PCM D100 (Sony, Tokyo, Japan). In terms of the thermal environment, air temperature (Ta) and relative humidity (RH) were measured for 5 min. The weather conditions were clear, and the wind speed was less than 5 m/s when measuring these physical parameters. All parameters were measured during people's activity times from morning (8:00–12:00 a.m.) to afternoon (2:00–6:00 p.m.).

2.3. Data Collection of Visual, Acoustic, and Thermal Evaluation

The questionnaire design referred to previous studies on the investigation of visual landscapes, soundscapes, and thermal environments, consisting of 4 parts, including demographic information, visual, acoustic, and thermal evaluation (Appendix A) [42,43]. In terms of demographic information, participants were required to list their age, gender, and educational level. Except for the above information, the subjective evaluation used a 5-point Likert scale. The visual evaluation part intended to explore the spatial openness and landscape elements perceived by people, including the percentage of natural elements and color richness. On the other hand, the soundscape evaluation investigated the soundscape pleasantness, tranquility, and comfort of the acoustic environment, while the soundscape pleasantness and tranquility were essential for acoustic comfort. As for thermal perception, the thermal sensitivity, sensitivity to solar radiation, and comfort vote were selected as the indicators.

For the visual landscape, previous studies indicated that the openness of natural green spaces, percentage of natural elements, and color richness significantly affected the landscape attraction and overall comfort [44]. Therefore, the subjective evaluation of the visual environment was conducted using spatial openness (from 1 = extremely enclosed to 5 = extremely open), landscape naturalness (from 1 = extremely low to 5 = extremely high), and color richness (from 1 = very homogeneous to 5 = very diverse).

For the soundscape evaluation, in order to analyze the subjective evaluation of the acoustic environment, the soundscape part of the questionnaire was designed following the suggestion of ISO/TS 12913-2:2018 [45]. Acoustic comfort was evaluated from the perspective of pleasantness (from 1= extremely annoyed to 5 = extremely pleasant), tranquility (from 1= extremely noisy to 5 = extremely tranquil), and comfort (from 1 = very uncomfortable to 5 = very comfortable).

Regarding the thermal evaluation, previous studies revealed that thermal sensitivity and sensitivity to solar radiation significantly affected thermal comfort. Therefore, the thermal evaluation was based on thermal sensitivity (from 1 = extremely hot to 5 = extremely cool), sensitivity to solar radiation (from 1 = very obvious to 5 = very sparse), and thermal comfort (from 1 = extremely uncomfortable to 5 = extremely comfortable). Overall, a principal component analysis (PCA) was performed to abstract indicators with high contributions to the visual landscape evaluation (VE), soundscape evaluation (AE), and thermal evaluation (TE). The scores of VE, AE, and TE were then calculated as the mean value of the above indicators. Finally, a total of 587 questionnaires were distributed at the six measuring points of the greenway plazas, and 554 valid questionnaires were collected. Participants were dominated by teenagers (15–24 years) and young people (25–65 years), at 37.5% and 49.8%. The number of females participating in the questionnaire was slightly more than the number of males (53.8% vs. 46.2%). Meanwhile, most participants had an educational level of high school, secondary school, and university, accounting for 80.9% of the total number.

2.4. Behavioral Observation

The users' behaviors on the greenway plaza were recorded with a hidden camera to prevent the influence of the recording process on the crowds' activities. The video recording was used to identify their behaviors using three variables, including the behavior type, frequency of users' behaviors, and proportion of duration time on their behaviors. All variables were calculated by Behavioral Observation Interactive Software (BORIS, v.2.95, University of Torino, Torino, Italy). Each recording lasted 20 min at each measuring plaza, divided into four segments. Once the behavior of a person lasted more than 10 s, the behavior type was determined based on the number of people, interpersonal distance (IPD), duration time, and activity purposes (Table 2). Single-person behaviors were identified as sitting, exercising, and taking photos based on posture and duration time. A one-to-one person behavior was noted if the IPD was 45–120 cm more than 10 s, while multi-person behavior was identified if the IPD was 129–365 cm more than 10 s based on posture [46]. The start and end times of each type of behavior were also recorded to determine the mean frequency and proportion of duration time of the behaviors.

Table 2. Identification of human behaviors as single-person, one-to-one person, and multi-person behaviors based on interpersonal distances and duration times.

Identification of Behavior	Activities Purposes	Interpersonal Distance	Duration Time
Single person behavior	Sitting/resting	/	T > 1 min
	Exercising	/	T > 1 min
	Walking	/	T > 1 min
	Taking photos/enjoying the view	/	T > 30 s
One-to-one person behavior	Chatting	45–120 cm	T > 10 s
	Face to face	45–120 cm	T > 10 s
	Walking when holding hands	0–45 cm	T > 10 s
Multi-person behavior	Playing near water	129–365 cm	T > 10 s
	Catching fish	129–365 cm	T > 10 s
	Group talking	129–365 cm	T > 10 s

2.5. Data Analysis

The experimental results were analyzed in two parts using Statistical Package for Social Scientists (SPSS 22.0, IBM). The first part aimed to identify the features of the visual, acoustic, and thermal environments with different spatial openness, as well as the subjective evaluation. A comparative analysis was further conducted to analyze the difference between those variables and their interactive effect. The second part aimed to explore the relationship between the interaction of the subjective evaluation and users' behaviors. These analyses helped us to identify the overall characteristics of the subjective evaluation and investigate the impact of those evaluations on their behaviors.

For the first part, descriptive analysis was used to calculate the mean and variance of the physical parameters and the visual, acoustic, and thermal evaluation ratings. Secondly, principal component analysis (PCA) was used to extract the main influencing factors of those subjective ratings. The mean value of landscape openness, landscape naturalness, color and richness; soundscape pleasantness, soundscape tranquility, and soundscape comfort; and thermal sensitivity, sensitivity to solar radiation, and thermal comfort were input into the PCA to decide and calculate the VE, AE, and TE. Any factor load that was below 0.6 was removed from the PCA. Then, the Kruskal–Wallis test was used to examine the significant differences between those influencing factors at each greenway plaza with different spatial openness. Based on the result of the one-way Kolmogorov–Smirnov test, Spearman's correlation coefficients r and p were used to identify the relationship between

visual, acoustic, thermal evaluation, and behavior indicators, where a p less than 0.05 represented significant results. For the second part, a multiple linear regression model was established to further explore the relationship between the subjective evaluation and users' behaviors. Adjusted R^2 and βcoefficients were used to assess the quality of the model. Variables meeting $p < 0.05$ and VIF < 2 were retained in the model.

3. Results

3.1. Characteristics of the Physical Environment and Multisensory Interactions at Greenway Plazas with Different Spatial Openness

3.1.1. Characteristics of the Physical Environment at Greenway Plazas with Different Spatial Openness

To answer RQ1, physical environment parameters, such as $L_{Aeq,5min}$, L10, L90, air temperature (Ta), and relative humidity (RH), were measured at each measuring point (Figure 3). The results indicated that the $L_{Aeq,5min}$ and L10 values of the enclosed plazas (R1 and R4) were higher than the other points (Figure 3a). As the distance away from the entrance of the greenway increased, both $L_{Aeq,5min}$ and L10 gradually dropped at each greenway plaza. This may be explained by the fact that the crowd density of R1 and R4 was higher than that of the other points, which contributed to a high sound pressure level due to human sounds, such as playing, talking, and laughing. On the other hand, R3 was dominated by natural sounds, such as birds, crickets, and water sounds, with the lowest L90 among all plazas. Although the open plazas (R5, R6) were farther from the entrance of the greenway than the enclosed and semi-enclosed plazas (R1, R2, R3, and R4), the $L_{Aeq,5min}$ of the open plazas was higher than R3, dominated by natural sounds because of the sounds of group activities. Regarding the thermal environment, the enclosed plazas (R1 and R4) had the lowest air temperature (Ta) but relatively highest humidity (RH) (Figure 3b,c). This indicated that the enclosure provided by the trees and shrubs of the enclosed plazas protected the thermal environment from solar radiation and created a cool environment in the plazas. Interestingly, the Ta values of the semi-enclosed plazas (R2 and R3) were higher than those of the open plazas (R5 and R6), while the RH values of the semi-enclosed plazas were lower than those of the open plazas. A possible explanation was that R2 and R3 had less vegetation than R5 and R6, which resulted in fluctuating temperatures in R2 and R3 during the hot summer climate.

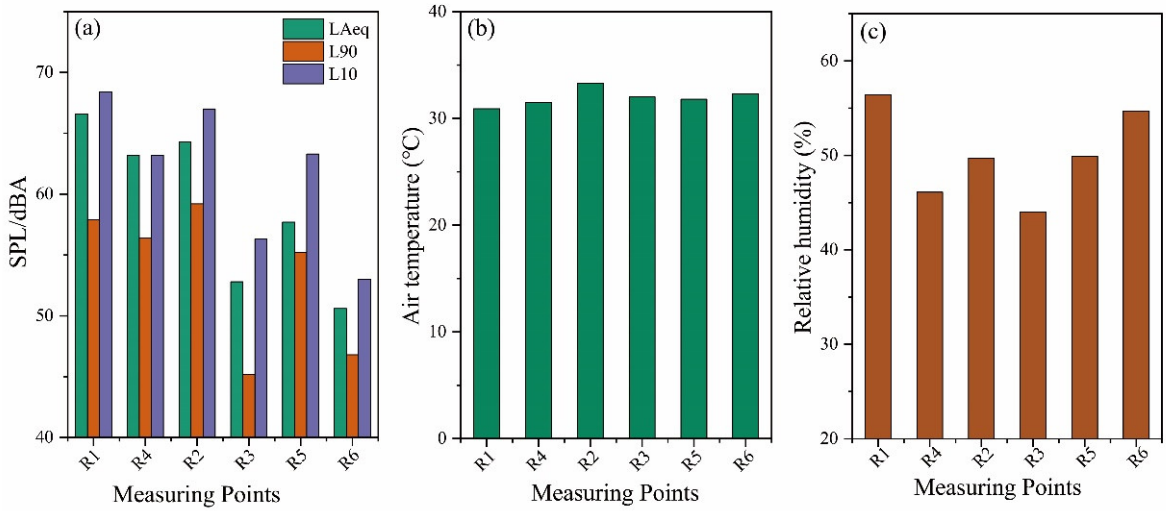


Figure 3. Mean value of visual, acoustic, and thermal environment parameters at each measuring point: (**a**) $L_{Aeq,5min}$, L10, and L90 (dBA); (**b**) air temperature (Ta); (**c**) relative humidity (RH).

3.1.2. Characteristics of the Subjective Evaluation of Greenway Plazas with Different Spatial Openness

To investigate the subjective evaluation of greenway plazas, we conducted a principal component analysis and the Kruskal–Wallis test to compare the visual, acoustic, and thermal perceptions between greenway plazas with different spatial openness: enclosed, semi-enclosed, and open plazas. Table 3 shows that landscape openness, soundscape tranquility, and thermal sensitivity significantly affected the subjective perception, with loading factors ranging from 0.782 to 0.813. The participants felt comfortable and satisfied with the visual, acoustic, and thermal environments, as the mean subjective ratings were above 3 (Figure 4). The VE ratings of the participants were higher than the AE and TE ratings at all plazas, indicating that people preferred landscape elements, such as natural vegetation, water bodies, and the sky of greenway plazas, more than the acoustic and thermal environments. Regarding the AE, the AE of the enclosed and open plazas was significantly higher than that of the semi-enclosed plazas ($p < 0.05$). This may be related to the fact that the enclosure of trees and shrubs in the enclosed plazas assisted in creating a tranquil space for people, while the artificial sounds in the semi-enclosed plazas created a noisy space and decreased the acoustic comfort. For the TE ratings, the TE of the open plazas was significantly lower than that of the enclosed and semi-enclosed plazas ($p < 0.05$). This suggested that people felt hotter in the open plazas under the impact of direct sunlight. However, no significant difference was found between the enclosed, semi-enclosed, and open plazas from VE perception ($p > 0.05$). This may be explained by the fact that natural vegetation was rich along the greenway, which made people comfortable in greenway plazas with all levels of spatial openness.

Table 3. Main factors representing visual, acoustic, and thermal evaluation from the questionnaire using principal component analysis (the indicators with a factor load < 0.6 were removed).

Descriptive Factors	Factors Load	Variance Contribution
Factor 1: Thermal Evaluation		24.2%
Thermal Sensitivity	0.813	
Sensitivity to Solar Radiation	0.782	
Thermal Comfort	0.760	
Factor 2: Visual Evaluation		43.5%
Landscape Openness	0.794	
Naturalness	0.723	
Soundscape Pleasantness	0.607	
Factor 3: Acoustic Evaluation		61.3%
Soundscape Tranquility	0.782	
Soundscape Comfort	0.768	

3.1.3. Characteristics of Multisensory Interactions at Greenway Plazas with Different Spatial Openness

Figure 5 illustrates the relationship between the visual, acoustic, and thermal evaluations in the greenway plazas with different levels of spatial openness using Spearman's correlation. As the diameter of the red circles becomes larger, the coefficient numbers become larger. Figure 5a–c all show a red circle for the correlation between VE and AE. This indicates that VE was positively correlated with AE ($p < 0.01$) in the greenway plazas with all levels of spatial openness. This suggests that a natural and comfortable visual landscape was beneficial for creating a quiet atmosphere in the greenway plazas. Meanwhile, the diameter of the red circle in (b) is larger than that in (a). This indicates that as the spatial openness increased, the correlation effect between VE and AE became more pronounced ($p < 0.001$), as well as the effect between VE and TE in the semi-enclosed plazas (R2, R3)

($p < 0.001$). This may be related to the fact that the semi-enclosed plazas were covered with an artificial landscape, while improving the proportion of green coverage was beneficial for increasing the visual evaluation, so as to improve thermal and acoustic comfort. However, AE was only positively correlated with TE in the enclosed plazas (R1, R4) ($p < 0.05$). This indicated that the quiet acoustic environment of the enclosed plazas assisted people in being less affected by high air temperatures.

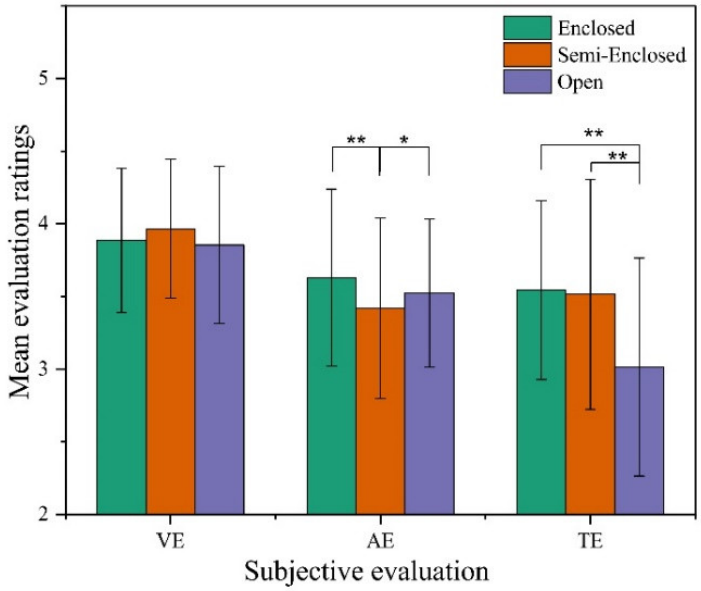


Figure 4. Mean value of each subjective evaluation (VE, AE, and TE) of each measuring point across different plaza types. Error bars indicate standard errors. (Note: VE = visual evaluation, AE = acoustic evaluation, TE = thermal evaluation; significant differences are marked as $p < 0.05$ * and $p < 0.01$ **).

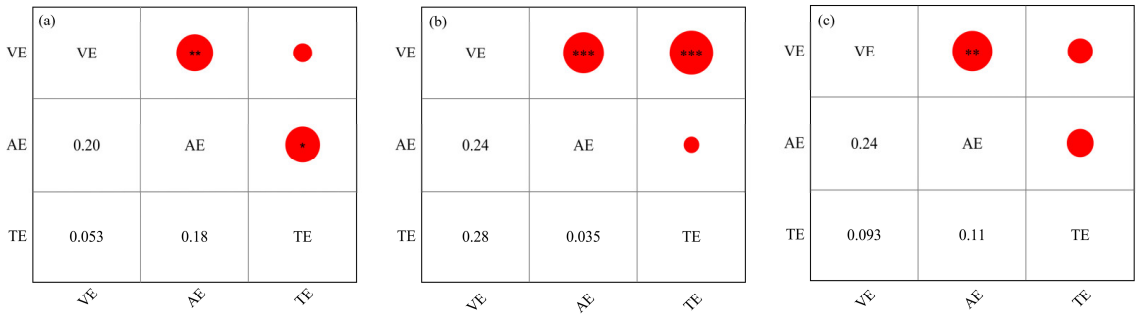


Figure 5. Spearman's correlation for the relationship between visual, acoustic, and thermal evaluations in greenway plazas with different spatial openness (significant correlations are marked with $p < 0.05$ *, $p < 0.01$ **, and $p < 0.001$ ***): (**a**) enclosed plaza, (**b**) semi-enclosed plaza, and (**c**) open plaza.

A multiple linear regression model was built to further explore the impact of spatial openness on the interactive effect of the subjective evaluation (Figure 6). Figure 6a demonstrates the regression line between the mean values of VE, TE, and AE. It indicates that the effect of AE on TE (TE = 0.042AE + 2.88) was stronger than that of AE on VE (VE = 0.03AE + 3.62). This shows that creating a tranquil environment is much more effective for improving thermal comfort in an enclosed plaza than increasing visual comfort during the hot summer climate. Figure 6b,c show the regression line between VE and AE as the spatial openness gradually increased. The correlation effect becomes more pronounced from the semi-enclosed plazas (AE = 0.064VE + 2.06) to the open plazas (AE = 0.085VE + 2.43). This proved the fact that a tranquil environment could be a significant factor in enhancing the landscape aesthetics for visitors in the open space of the urban greenway. However, there is no significant correlation between AE and TE in the

semi-enclosed and open plazas ($p > 0.05$). This may be related to the fact that the solar radiation of open plazas significantly influenced the thermal evaluation more than the impact of the acoustic environment. This result provided further answers to RQ1.

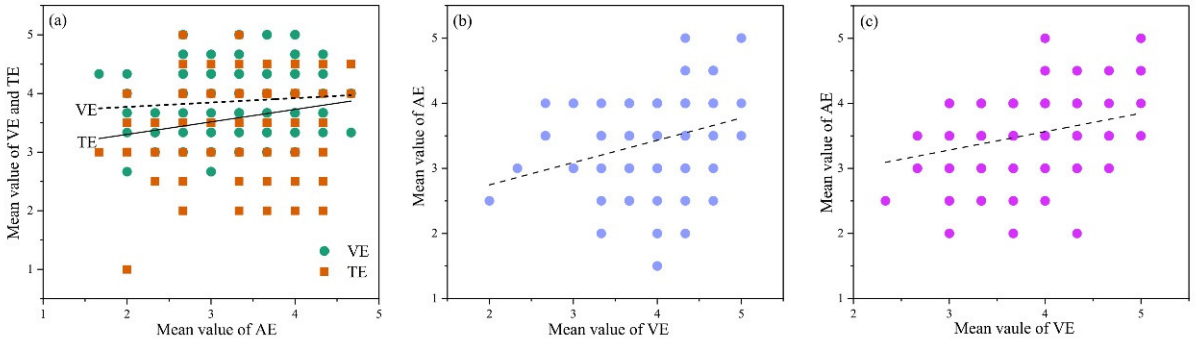


Figure 6. Linear regression curve between visual, acoustic, and thermal evaluations of greenway plazas with different spatial openness: (**a**) AE, VE, and TE of the enclosed plazas; (**b**) AE and VE of the semi-enclosed plazas; (**c**) AE and VE of the open plazas (note: AE: acoustic evaluation; TE: thermal evaluation; VE: visual evaluation). The dashed line represents the regression line between each indicator.

3.2. Effect of Multisensory Interactions on Users' Behaviors at Greenway Plazas with Different Levels of Spatial Openness

3.2.1. Characteristics of Users' Behaviors at Greenway Plazas with Different Levels of Spatial Openness

Figure 7 illustrates the mean frequency and duration time of different types of behaviors at the greenway plazas with each level of spatial openness. For the frequency of users' behaviors, the frequency of single-person behavior (FSB) and one-to-one person behavior (FOB) were both higher than that of multi-person behavior (FMB) in the enclosed and semi-enclosed greenway plazas (Figure 7a). This indicated that people tended to perform single- or two-person activities, such as staying, taking photos, and having conversations in plazas with enclosures along the greenway. As the spatial openness increased, the FMB gradually increased and reached a maximum value in the open plazas (open: 2.71 vs. enclosed: 0.67, semi-enclosed: 2.0; $p > 0.05$) compared to FSB and FOB. This suggested that plazas with a broad view motivated people to carry out group activities, such as playing near water and catching fish. In terms of the duration time of users' behaviors, the time of both single-person behavior (SPB) and one-to-one person behavior (OPB) decreased as the spatial openness increased (Figure 7b). This showed that people prefer activities with one or two people in enclosed plazas with a quiet atmosphere. Conversely, the duration time of multi-person behavior (MPB) was higher in the semi-enclosed and open plazas than that in the enclosed plazas (semi-enclosed: 39.1%, open: 39.6% vs. enclosed: 16.7%; $p > 0.05$). This proved that plazas with high spatial openness were beneficial not only for the number but also the time that people were willing to spend on group activities. Interestingly, the duration time of OPB in the open plazas was the lowest one among all plazas. This suggested that open plazas lacked private spaces for two-person social interactions.

3.2.2. Effect of Multisensory Interaction on Users' Behaviors Across Different Types of Plazas

Table 4 shows the effect of the subjective evaluation on users' behaviors from visual, acoustic, and thermal perceptions. First, regarding the thermal–acoustic evaluation interaction, it was found that both AE and TE had a positive impact on FOB at the enclosed and semi-enclosed plazas ($p < 0.05$). This demonstrated that a tranquil and cool environment of greenway plazas with enclosures was suitable for people to stay and have conversa-

tions. Conversely, in terms of the audio-visual evaluation interaction, both VE and AE were negatively correlated with OPB and MPB, respectively ($p < 0.05$). This meant that people preferred artificial sounds with high sound pressure levels to immerse themselves in interactive activities. However, no significant correlation was found between the subjective evaluation and users' behaviors in the open plazas ($p > 0.05$). That may be related to the fact that direct sun radiation prevented people from performing outdoor activities in the open plazas.

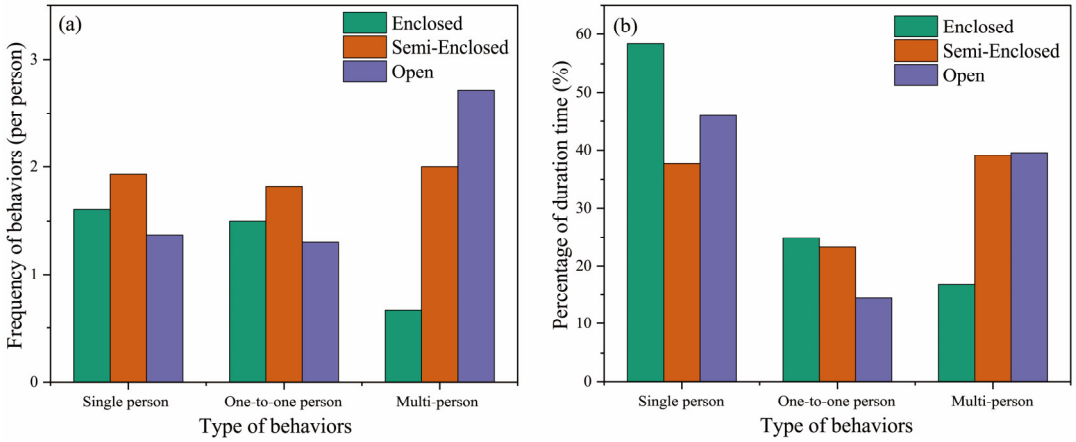


Figure 7. Indicators of users' behaviors across greenway plazas with different levels of spatial openness: (**a**) mean frequency of users' behaviors (times/per person); (**b**) proportion of duration time of users' behaviors (%).

Table 4. Spearman's rho correlation coefficients for the relationship between visual, acoustic, and thermal evaluations and users' behaviors in the greenway plazas ($p < 0.05$ *).

Spatial Openness		VE	AE	TE
Enclosed Plaza	FOB	0.355	0.753 *	0.355
Semi-enclosed Plaza	FOB	0.121	−0.388	0.597 *
	OPB/%	−0.574 *	0.014	−0.159
	MPB/%	0.308	−0.610 *	−0.063

A multiple linear regression model was also further constructed to explore the relationship between the subjective evaluation and users' behaviors under different spatial openness conditions (Table 5). Regarding VE and AE, the impact of AE on FTB ($AdjR^2 = 0.696$, $\beta = 0.744$, $p < 0.01$) was stronger than that of TE on FOB ($AdjR^2 = 0.696$, $\beta = 0.568$, $p < 0.01$) in the enclosed plazas, while VE had no significant impact on FOB ($p > 0.05$). This proved that AE was the key factor for motivating people to engage in interactive activities in greenway plazas with high enclosures. However, the impact of AE on FOB gradually became weakened ($p > 0.05$) as the spatial openness increased. This suggested that VE and AE didn't significantly affect users' behavior in the open plaza. For TE, TE had a more significant positive impact on FOB ($AdjR^2 = 0.340$, $\beta = 0.575$) in the semi-enclosed plazas than AE did. This proved that creating a cool environment was effective for increasing the frequency of two-person social interactions. This provided answers to RQ2.

Table 5. Results of the multi-linear regression analysis between subjective evaluation from visual, acoustic, and thermal perceptions and users' behavior of the greenway plazas.

Spatial Openness	Dependent Variable	Independent Variable	Adj R^2	β	*p*
Enclosed Plaza	FOB	AE	0.696	0.744	0.005
		TE		0.568	0.018
Semi-enclosed Plaza	FOB	TE	0.340	0.575	0.016

4. Discussion

4.1. Interactive Effect Between Spatial Openness and Single Subjective Evaluation in Greenway Plazas

This study indicated that spatial openness had a negative impact on the thermal evaluation ($p < 0.05$), while the visual evaluation was less influenced by spatial openness ($p > 0.05$). Regarding the acoustic environment, the sound of human activities affected the acoustic evaluation more than spatial openness did. From the perspective of the visual evaluation, previous studies revealed that an increase in spatial openness led to a greater proportion of natural elements, such as water bodies and vegetation cover, which further promoted landscape aesthetics and positive emotions [47]. This contradicts our study result. This may be related to the fact that the enclosed structure along the urban greenway was green coverage, such as trees and shrubs, which could also be the natural elements that make people feel pleasant. Regarding the acoustic environment, our study found that the acoustic evaluation results of the enclosed plazas close to the greenway entrance were rated more comfortable than those of the other plazas ($p < 0.01$), even with high sound pressure levels. Conversely, other studies revealed that traffic noise with a high sound pressure level had a detrimental effect on noise sensitivity in the natural view of parks, leading to anxiety and negative overall comfort [48–50]. A possible explanation for the difference was that the coverage of trees and green vegetation in greenway plazas reduced the noise annoyance of visitors and increased the acoustic evaluation. In terms of the thermal evaluation, our study results suggested that the enclosed plazas significantly increased the thermal evaluation compared to the open plazas ($p < 0.01$). However, previous studies revealed that people felt less comfortable in the thermal environment in enclosed spaces surrounded by grey infrastructure along a greenway [51]. This may be explained by the fact that the greenway plazas selected in our study were near water bodies and green vegetation instead of grey infrastructures, which reduced the temperature fluctuation and contributed to thermal comfort.

4.2. Interactive Effect Between Spatial Openness and Multisensory Interactions in Greenway Plazas

Our study results indicated that spatial openness had a positive effect on the interactive effect between visual and acoustic evaluations. Previous studies also found the same result that an open landscape view, such as water bodies matched with forests, activated positive emotions more easily than enclosed views, such as mountains [18,27]. Those positive emotions were beneficial for visual and acoustic evaluations. Our study also found that the acoustic evaluation was the key factor influencing the visual and thermal perspectives. Unlike this result, previous studies revealed that visual and thermal evaluations affected overall satisfaction more than an acoustic evaluation did, while visual landscape comfort significantly influenced soundscape pleasantness in coastal greenways and parks [52,53]. This may be explained by the fact that the continuous green coverage and enclosures along the linear greenway made people keep the same feeling of the visual landscape, while the changing acoustic environment, such as natural sounds changing to human sounds, affected the way they perceived the physical environment.

On the other hand, a significant correlation between the visual and thermal evaluations was observed only in the semi-enclosed plazas surrounded by grey infrastructures ($p < 0.01$). Previous studies also found that both visual and thermal evaluations had positive impacts on overall comfort [54,55]. This proved that increasing the green coverage of semi-enclosed plazas could be beneficial for overall comfort. However, a study on a cold winter park revealed that a visual evaluation had a more significant effect on acoustic perception than a thermal evaluation did [38]. This contradicts our study result. One possible explanation

for this was that unlike the winter park, the summer greenway plazas were rich in green coverage, which emphasized the role of the thermal evaluation.

For thermal–acoustic perception, our study revealed a weak correlation between the thermal and acoustic evaluations in the open plazas. Conversely, other studies found that intensified green vegetation created a more positive effect between thermal–acoustic perception and psychological restoration compared to sparse vegetation in urban greenways, while thermal acceptability directly affected soundscape assessments in communities' open spaces [30,52,56]. This may be related to the fact that the percentage of green vegetation of enclosed plazas was more than that of open plazas in communities' open spaces, which strengthened the correlation effect between thermal and acoustic perceptions.

4.3. Interactive Effect Between Spatial Openness and Users' Behaviors in Greenway Plazas

The thermal–acoustic evaluation had a positive effect on the frequency of interactive behavior in the semi-enclosed and enclosed plazas, while the audio-visual evaluation was negatively associated with the duration time of those behaviors only in the semi-enclosed plazas. From the perspective of the frequency of users' behaviors, previous studies found contradictory results that the enclosed space of urban greenways, rather than the acoustic environment, encouraged many cycling behaviors, with transportation infrastructures and greenery coverage enhancing this effect [57,58]. This may be related to the fact that interactive activities were relatively stationary at plazas, which allowed people to immerse themselves in the acoustic and thermal environments, while cycling sports with moving positions were more affected by the surrounding landscape. Another finding of our study was that the acoustic evaluation affected the one-to-one interactive behavior more than the thermal evaluation did. Previous studies had similar results, suggesting that interactive activities, such as talking, needed a tranquil and private space, while the enclosure of the tree canopy allowed people to be less affected by sun radiation [59,60]. For the duration time of users' behaviors, other studies on the usage of greenways indicated that nature-based landscape views, such as waterfront spaces, motivated people to stay and spend time in outdoor activities, while people who preferred sports exercises were more willing to stay in public facility-based landscapes coupled with activity sounds [36,37,61]. These findings proved our study result that an artificial landscape, such as public facilities, should be added in semi-enclosed plazas to satisfy the needs of people who want to carry out physical activities.

4.4. Suggestions for Promoting Subjective Evaluation and Interactive Activities in Urban Greenways

According to our study results, the spatial openness of plazas affects acoustic and thermal evaluations, leading to different sensory interactions and activities. Therefore, each level of spatial openness of plazas should be applied with corresponding strategies to satisfy restorative and activity needs. For enclosed plazas, considering the impact of the acoustic evaluation on other perspectives of evaluation, a tranquil environment needs to be regarded as a priority to be created, such as controlling the speed of bikes and honking times, to improve the overall satisfaction. This strategy was also beneficial for one-to-one social interaction, such as talking and enjoying the view, according to our study results. Regarding semi-enclosed plazas, the physical environment should be designed based on the functions of plazas. Firstly, increasing the percentage of green coverage, such as plants and shrubs, could effectively improve the visual and thermal evaluations and purify hazardous matter based on the correlation result. Secondly, designing more leisure space to attract people to stay and play could increase the sound pressure level of human activity sounds, thus enhancing the immersive feeling of people and stimulating multi-person interactions, such as catching fish. In terms of open plazas, a tranquil space could improve the visual

evaluation using strategies such as turning down the volume of music in the nearby store. Since the thermal evaluation was damaged by solar radiation, more trees are needed to create sun shades and provide spaces for multi-person activities, like playing near water.

On the other hand, our study result has limitations related to the fact that most of the participants were teenagers and young people. Children and elderly people may have other types of multisensory evaluations and behaviors at greenway plazas, such as playing and sitting for longer times than young groups. And educational backgrounds may also influence people's behavior choices. In other ways, other landscape elements, such as vegetation coverage, enclosure of grey structures, and shapes of shorelines, could also influence the spatial openness and multisensory evaluations. In the future, more age groups, educational groups, and landscape elements will be taken into account to improve the generalization of strategies for city planners.

5. Conclusions

In the context of rapid restoration needs and the eco-function of urban greenways, this study examined the effect of spatial openness on multisensory interactions and users' behaviors in greenway plazas and provided corresponding strategies for creating eco-friendly environments and enhancing the well-being of people in urban greenways. Our study results revealed that spatial openness had a negative impact on the thermal evaluation, while people felt annoyed by the high level of artificial sounds in the semi-enclosed plazas rather than being influenced by spatial openness. Secondly, spatial openness had a positive effect on the interactions between visual and acoustic perceptions, while the visual and thermal evaluations were only correlated in the semi-enclosed plazas. Thirdly, the acoustic evaluations should be improved to enhance one-to-one social interactions in the enclosed plazas, while enhancing the thermal evaluation was beneficial for one-to-one social interactions in the semi-enclosed plazas. Based on the above results, different levels of spatial openness of greenway plazas were applied, with corresponding strategies such as controlling traffic noise, increasing green coverage, and designing more leisure spaces. These strategies not only improve the multisensory evaluation, but they also improve the air quality and are beneficial for users' health. Future studies need to include larger numbers of children and elder groups to enhance the generalization of conclusions and consider more types of evaluations, such as smell and touch senses.

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Institutional Review Board Statement: This study was conducted in accordance with the Declaration of Helsinki. Formal ethical approval was waived by Huazhong Agricultural University for this research, as it involved an anonymous questionnaire with minimal risk, no collection of personally identifiable information, and no impact on participants' legal rights or welfare. All data were anonymized and remained confidential during the analysis.

Informed Consent Statement: Informed consent was obtained from all participants before they participated in the study.

Data Availability Statement: Data will be made available on request.

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Abbreviations

The following abbreviations are used in this manuscript:

Ta	Air temperature
RH	Relative humidity
VE	Visual evaluation
AE	Acoustic evaluation
TE	Thermal evaluation
FSB	Frequency of single-person behavior
FOB	Frequency of one-to-one person behavior
FMB	Frequency of multi-person behavior
SPB	Duration time of single-person behavior
OPB	Duration time of one-to-one person behavior
MPB	Duration time of multi-person behavior

Appendix A. Questionnaire of This Study

Demographic Information	
Age	☐ 8–14 years ☐ 15–24 years ☐ 25–65 years ☐ above 65 years
Gender	☐ Male ☐ Female ☐ More
Educational Level	☐ Primary school and others ☐ Middle school ☐ High/secondary school ☐ Universities ☐ Post graduate and above
Visual evaluation	
What do you think of the spatial openness of the current plaza?	☐ Totally enclosed~ ☐ Totally open (1~5)
What do you think of the percentage of natural scene such as sky, water bodies and trees in your vision at the current plaza?	☐ Extremely low~ ☐ Extremely high (1~5)
What do you think of the plant colors around the current plaza?	☐ Very single~ ☐ Very diversified (1~5)
Acoustic evaluation	
How do you rate the pleasantness of the acoustic environment at the current plaza?	☐ Very annoying~ ☐ Very pleasant (1~5)
What do you think of the tranquility of current plaza?	☐ Very noisy~ ☐ Very tranquil (1~5)
What do you think of the overall comfort of the acoustic environment at current plaza?	☐ Very discomfort~ ☐ Very comfort (1~5)
Thermal evaluation	
What do you feel on the air temperature of the current plaza?	☐ Very hot~ ☐ Very cool (1~5)
How do you feel about the degree of direct sunlight at the current plaza?	☐ Very obvious~ ☐ Very scarce (1~5)
How do you rate on the overall comfort of thermal environment at current plaza?	☐ Very discomfort~ ☐ Very comfort (1~5)

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